

TalentGeenie — Software Requirements Specification (SRS)

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Product: TalentGeenie — AI-Powered Interview & Learning Platform
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Table of Contents

- 1. [Introduction](#)
 - 2. [Overall Description](#)
 - 3. [System Features & Functional Requirements](#)
 - 4. [External Interface Requirements](#)
 - 5. [Non-Functional Requirements](#)
 - 6. [Data Requirements](#)
 - 7. [Security Requirements](#)
 - 8. [Appendices](#)
-

1. Introduction

1.1 Purpose

This document specifies the complete software requirements for **TalentGeenie**, a multi-tenant, AI-powered interview management and learning platform. It serves as the single source of truth for development, testing, and acceptance criteria.

1.2 Scope

TalentGeenie is an end-to-end platform that enables organizations to:

- Create and manage AI-generated interview assessments
- Evaluate candidates with AI-powered scoring and hiring recommendations
- Proctor interviews with video, audio, tab-switch, face, and gaze detection
- Manage employee learning paths, certifications, and training
- Handle multi-organization billing with subscription plans
- Provide role-based hierarchical access for 6 distinct user types

1.3 Definitions & Acronyms

Term	Definition
JD	Job Description
CPI	Candidate Performance Index — composite score across technical, problem-solving, communication, and integrity dimensions
RBAC	Role-Based Access Control
RLS	Row-Level Security (PostgreSQL)

MCQ	Multiple-Choice Question
ATS	Applicant Tracking System
SPA	Single-Page Application
Edge Function	Serverless Deno function deployed on Supabase

1.4 Intended Audience

- **Developers/Engineers** — Implementation reference
- **QA/Test Engineers** — Acceptance criteria and test planning
- **Product Owners** — Feature validation and prioritization
- **Stakeholders** — Business requirement verification

1.5 References

Document	Location
High-Level Design (HLD)	docs/HIGH_LEVEL_DESIGN.md
Low-Level Design (LLD)	docs/LOW_LEVEL_DESIGN.md
Security Audit Report	SECURITY_AUDIT_COMPLETE.md
Database Migration Scripts	docs/COMPLETE_DATABASE_MIGRATIONS.md
Deployment Guide	DEPLOYMENT.md

2. Overall Description

2.1 Product Perspective

TalentGeenie is a **standalone SaaS platform** that integrates:

```
flowchart TD
    subgraph Platform["TalentGeenie Platform"]
        IM["Interview<br/>Management"]
        LC["Learning &<br/>Certifications"]
        PE["Proctoring<br/>Engine"]
        BS["Billing &<br/>Subscriptions"]

        IM --> AI
        LC --> AI
        PE --> AI
        BS --> AI

        AI["AI Engine<br/>(Gemini 2.5 Flash)"]
        AI --> Backend

        Backend["Self-Hosted Supabase Backend<br/>PostgreSQL · PostgREST · GoTrue · Kong · Storage"]
    end
```

end

```
style Platform fill:#f0f4ff,stroke:#3b82f6,stroke-width:2px
style AI fill:#fef3c7,stroke:#f59e0b,stroke-width:2px
style Backend fill:#d1fae5,stroke:#10b981,stroke-width:2px
```

2.2 Product Functions (Summary)

#	Function Area	Description
F1	Interview Management	Create, configure, and manage AI-generated interview templates with customizable question types, difficulty, and topics
F2	AI Question Generation	Generate tailored interview questions from job descriptions using Google Gemini AI with configurable type/difficulty distributions
F3	Candidate Assessment	AI-evaluate candidate responses with overall scores, topic-wise breakdown, hiring recommendations, strengths/weaknesses
F4	Proctoring	Real-time video/audio proctoring with 17+ violation types, face detection, gaze tracking, VM detection, and integrity scoring
F5	Learning & Training	Role-based training plans with topic-wise progress tracking, practice assessments, and certification preparation
F6	Certifications	Proctored certification exams (Azure, AWS, GCP, etc.) with certificate issuance and public verification
F7	Multi-Tenant Organizations	Organization management with isolated data, configurable settings, and member management
F8	Subscription & Billing	Plan-based billing with Stripe integration, invoice generation, usage tracking, and payment management
F9	RBAC & Security	6-role hierarchical access control with 38 granular permissions, RLS policies, and audit logging
F10	Analytics & Reporting	Interview analytics, organization metrics, AI usage monitoring, and exportable reports
F11	JD Builder Wizard	AI-assisted job description creation with skill extraction and template suggestions
F12	ATS Integration	Integration with external Applicant Tracking Systems via webhooks

2.3 User Classes and Roles

flowchart TD

PA["🔑 Platform Admin
<i>GOD MODE – Full platform access</i>
All orgs, all data, all settings"]

PA --> PtA["🏢 Partner Admin
<i>Organization-level administration</i>
Own org settings, users, billing, recruiting"]

PtA --> HR["👤 HR Recruiter
Create/manage interviews,
evaluate, share,

```

proctor"]
  PtA --> TS["🏢 Tech SPOC<br/>Question contrib.,<br/>Templates, read-only
interviews"]
  PtA --> BC["💰 Billing Contact<br/>Invoices, payments,<br/>subscriptions"]
  PtA --> G["🎓 Guest<br/>Learning only,<br/>awaiting role assignment"]

  C["👤 Candidate<br/><i>External (unauthenticated)</i><br/>Takes interview via
shared link"]

style PA fill:#fee2e2,stroke:#ef4444,stroke-width:2px
style PtA fill:#fef3c7,stroke:#f59e0b,stroke-width:2px
style HR fill:#dbeafe,stroke:#3b82f6,stroke-width:2px
style TS fill:#dbeafe,stroke:#3b82f6,stroke-width:2px
style BC fill:#dbeafe,stroke:#3b82f6,stroke-width:2px
style G fill:#f3e8ff,stroke:#a855f7,stroke-width:2px
style C fill:#d1fae5,stroke:#10b981,stroke-width:2px

```

Role Permissions Matrix

Permission	Platform Admin	Partner Admin	HR Recruiter	Tech SPOC	Billing Contact	Guest
Interviews						
Create interviews	✓	✓	✓	✓	✗	✗
Edit interviews	✓	✓	✓	✓	✗	✗
Delete interviews	✓	✓	✗	✗	✗	✗
Share interviews	✓	✓	✓	✗	✗	✗
View interviews	✓	✓	✓	✓	✗	✗
Manage questions	✓	✓	✓	✓	✗	✗
Users						
Create users	✓	✓	✗	✗	✗	✗
Edit users	✓	✓	✗	✗	✗	✗
Delete users	✓	✗	✗	✗	✗	✗
View users	✓	✓	✓	✓	✗	✗
Assign roles	✓	✓	✗	✗	✗	✗
Manage teams	✓	✓	✗	✗	✗	✗
Reports						
View reports	✓	✓	✓	✓	✗	✗
Generate reports	✓	✓	✓	✗	✗	✗

Export reports	✓	✓	✓	✗	✗	✗
View assessments	✓	✓	✓	✓	✗	✗
Analytics						
View analytics	✓	✓	✓	✓	✗	✗
Org analytics	✓	✓	✗	✗	✗	✗
Comparative analytics	✓	✗	✗	✗	✗	✗
Billing						
View billing	✓	✓	✗	✗	✓	✗
Manage billing	✓	✓	✗	✗	✓	✗
Manage subscription	✓	✓	✗	✗	✓	✗
View invoices	✓	✓	✗	✗	✓	✗
Organizations						
Manage organizations	✓	✗	✗	✗	✗	✗
Approve applications	✓	✗	✗	✗	✗	✗
View org members	✓	✓	✗	✗	✗	✗
Settings						
Manage settings	✓	✓	✗	✗	✗	✗
Manage integrations	✓	✓	✗	✗	✗	✗
Manage custom roles	✓	✓	✗	✗	✗	✗
Proctoring						
View proctoring	✓	✓	✓	✗	✗	✗
Manage proctoring	✓	✓	✗	✗	✗	✗
Review violations	✓	✓	✓	✗	✗	✗
Learning						

View learning	✓	✓	✗	✓	✗	✗
Create learning	✓	✗	✗	✗	✗	✗
Manage learning	✓	✗	✗	✗	✗	✗

2.4 Operating Environment

Component	Requirement
Client Browser	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Mobile	iOS 14+ / Android 10+ (via Capacitor hybrid app)
Server OS	Linux (Docker containers)
Database	PostgreSQL 15.8+
Runtime	Node.js 20+, Deno (edge functions)
Container	Docker 24+, Kubernetes 1.28+ (optional)

2.5 Design and Implementation Constraints

- **C1:** Must use self-hosted Supabase (PostgreSQL, GoTrue, PostgREST, Kong)
- **C2:** AI provider must support Google Gemini 2.5 Flash (with 1.5 Flash fallback)
- **C3:** All database tables must have Row-Level Security (RLS) enabled
- **C4:** Frontend must be a React SPA deployable via Nginx
- **C5:** Must support multi-tenant organization isolation
- **C6:** Proctoring must work with standard browser MediaDevices API

2.6 Assumptions and Dependencies

- **A1:** Users have modern browsers with WebRTC support for proctoring
- **A2:** Candidates have camera + microphone for proctored interviews
- **A3:** Google Gemini API is available with valid API credentials
- **A4:** Stripe API is used for payment processing
- **A5:** Minimum 1 Mbps network bandwidth for proctored sessions

3. System Features & Functional Requirements

3.1 Authentication & User Management

FR-AUTH-01: User Registration

- **Description:** New users register with email, password, and full name
- **Validation Rules:**
 - Email: Valid format, ≤255 characters, trimmed and lowercased
 - Password: 8–72 characters, ≥1 uppercase, ≥1 lowercase, ≥1 digit
 - Full Name: 2–100 characters, letters/spaces/hyphens/apostrophes only (regex: `/^[a-zA-Z\s'–]+$/`)
- **Post-Registration:**
 - Profile created in `profiles` table with `email_verified: false`
 - Default `guest` role assigned via `user_roles` table

- Onboarding progress initialized
 - Welcome email sent
- **Priority:** Critical

FR-AUTH-02: Email Verification

- **Description:** Users must verify email before accessing protected features
- **Flow:** Registration → verification email → click link → redirect to `/auth/verify` → `email_verified` set to `true`
- **Bypass:** Platform admin can skip verification
- **Priority:** Critical

FR-AUTH-03: Sign In

- **Description:** Existing users sign in with email and password
- **Flow:** Enter email → Check user exists (RPC) → Route to sign-in → Enter password → Authenticate via GoTrue → Redirect to role-based dashboard
- **Dashboard Redirects:**
 - `platform_admin` → `/admin`
 - `partner_admin` → `/partner/portal`
 - `hr_recruiter` → `/partner/recruiting/interviews`
 - `tech_spoc` → `/partner/recruiting/question-repository`
 - `billing_contact` → `/partner/billing`
 - `guest` → `/learning-dashboard`
- **Priority:** Critical

FR-AUTH-04: Password Reset

- **Description:** Users can request password reset via email
- **Flow:** Enter email at `/reset-password` → reset email sent → click link → `/reset-password/confirm` → set new password
- **Priority:** High

FR-AUTH-05: Session Management

- **Description:** JWT-based sessions with automatic refresh
- **Details:**
 - Token expiration check with 60-second buffer
 - Automatic token refresh on expiration
 - Session stored securely in browser
- **Priority:** Critical

FR-AUTH-06: User Profile Management

- **Description:** All users can view and edit their profile
- **Editable Fields:** Full name, phone, bio, avatar
- **Validation:** Same rules as registration for name fields
- **Password Change:** Requires current password, validates strength rules, confirms match
- **Priority:** High

3.2 Interview Management

FR-INT-01: Interview Creation

- **Description:** HR Recruiters and above can create interview templates
- **Required Fields:**

- Title: 5–200 characters
- Job Description: 50–10,000 characters
- Question Count: 5–100 (integer)
- Time Limit: 0–180 minutes (optional)
- **Question Type Configuration:**
 - Types: MCQ, Scenario-Based, Coding, Descriptive
 - Counts per type (must sum to total question count)
 - Difficulty per type: Easy/Medium/Hard percentages (sum to 100%)
- **Topic Distribution:**
 - Skills extracted from JD via AI
 - Percentage per topic (must sum to 100%)
 - Required question rules (e.g., "at least 2 hard coding on SQL")
- **Additional Options:**
 - Proctoring enable/disable toggle
 - Question bank size (default: 200)
 - Experience level detection
- **Modes:** Create new, edit existing, reconfigure, pre-fill from JD Builder/template
- **Priority:** Critical

FR-INT-02: AI Question Generation

- **Description:** Generate interview questions from configured parameters
- **AI Provider:** Google Gemini 2.5 Flash (primary), 1.5 Flash (fallback)
- **Input:** Job description, question type distribution, difficulty distribution, topic distribution, required rules
- **Output:** Question bank with text, type, difficulty, topic, options (MCQ), correct answers, code templates (coding), test cases, explanations
- **Progress:** Real-time generation progress display
- **Post-Generation:** Questions stored in `questions` table linked to interview
- **Priority:** Critical

FR-INT-03: Interview Sharing

- **Description:** Share interviews with candidates via unique links
- **URL Formats:**
 - New: `/i/:orgSlug/:interviewSlug/:shareToken`
 - Legacy: `/take-interview/:shareLink`
- **Invitations:** Send via email with candidate name/email, expiry date
- **Token:** Cryptographically generated share token
- **Priority:** Critical

FR-INT-04: Candidate Interview Taking

- **Description:** Candidates take interviews via shared links (no account required)
- **Flow:**
 1. Resolve invitation via share token (3 retries, 1.5s → backoff)
 2. Validate: not expired, not completed, not archived
 3. Candidate enters name + email (must match invitation)
 4. Pre-interview proctoring checks (if enabled)
 5. Create attempt via secure RPC (generates 64-char session token)
 6. Fetch questions (correct answers hidden via RLS)
 7. Timer countdown with auto-submit on expiry

- 8. Submit answers with session token validation
- 9. Auto-evaluate trigger on submission

- **Validation:**
 - Candidate name: 2–100 characters, letters only
 - Candidate email: valid format, ≤255 characters
 - Session token: ≥32 characters, alphanumeric
 - Answer: 1–5,000 characters per question
 - Time taken: 0–86,400 seconds (24 hours max)
- **One attempt per candidate email per interview**
- **Priority:** Critical

FR-INT-05: AI Candidate Evaluation

- **Description:** Automated evaluation of candidate responses
- **Output:**
 - Overall score (0–100)
 - Hiring decision (Strong Hire / Hire / Lean Hire / No Hire)
 - Per-topic scores
 - Strengths and weaknesses arrays
 - Detailed analysis per question
 - CPI: Technical, Problem-Solving, Communication, Integrity scores
- **Trigger:** Automatic post-submission via edge function
- **Priority:** Critical

FR-INT-06: Interview List & Search

- **Description:** View all interviews with filtering and search
- **View Modes:** Card, Compact, Table
- **Filters:** Status (active, archived, draft), search by title/job
- **Priority:** High

FR-INT-07: Interview Detail View

- **Description:** Detailed view of interview configuration, questions, attempts, and assessment reports
 - **Priority:** High
-

3.3 Proctoring System

FR-PROC-01: Pre-Interview Checks

- **Description:** 4-step wizard before proctored interviews
- **Steps:**
 1. **Instructions:** Rules and guidelines display
 2. **Consent:** Must agree to proctoring, recording, data usage, AI analysis
 3. **Setup:** Automated hardware checks
 4. **Ready:** Confirmation to proceed
- **Automated Checks:**
 - Network bandwidth: 5 × 1MB samples, median calculation (fail < 1 Mbps, warn < 5 Mbps)
 - Camera: 640×480 MediaDevices validation
 - Microphone: Audio track presence on stream
 - Lighting: Camera quality + brightness analysis
 - Person visible: Face detection (single person required)
- **Priority:** Critical

FR-PROC-02: Real-Time Monitoring

- **Description:** Continuous monitoring during proctored assessments
- **Live Checks During Assessment:**
 - Tab switch detection
 - Print screen blocking
 - Multi-monitor detection
 - VM/virtual machine detection (single check)
 - Copy/paste attempt detection
 - Typing pattern analysis
 - Screen share continuity enforcement
- **Deferred Heavy Analysis (Post-Interview):**
 - Multi-person detection (face detection)
 - Gaze tracking and eye movement analysis
 - Voice analysis (multiple voices, playback)
 - Liveness detection
 - Face verification (person swap)
 - Phone/prohibited object detection
- **Priority:** Critical

FR-PROC-03: Violation Tracking

- **Description:** System tracks 17+ violation types with severity levels
- **Violation Types:**

Violation	Severity	Detection Method
Multiple persons	High	Face detection (post-interview)
Multiple voices	Medium	Voice analysis (post-interview)
Tab switch	Medium	Visibility change API
Look away	Low	Gaze tracking (>3–5s threshold)
Copy attempt	Medium	Clipboard event interception
Eye movement anomaly	Low	MediaPipe gaze tracking
Liveness failure	High	Liveness detection model
Person swap	High	Face verification mismatch
Silence anomaly	Low	Audio analysis
Phone detected	High	Object detection
Prohibited object	High	Object detection
Multiple monitors	Medium	Screen enumeration API
Print screen	High	Keyboard event interception
Virtual machine	High	Browser fingerprinting
Suspicious typing	Medium	Typing pattern analysis

Audio playback	Medium	Audio spectral analysis
Screen share stopped	High	MediaStream track ended

- **Priority:** Critical

FR-PROC-04: Recording Management

- **Description:** Video and screen recording during proctored interviews
- **Features:**
 - Chunked upload during interview (prevents data loss)
 - Post-interview chunk merging
 - WebM metadata repair
 - Signed upload URLs for security
 - Background upload with progress tracking
 - Stream health monitoring with auto-recovery
- **Priority:** High

FR-PROC-05: Integrity Scoring

- **Description:** Composite integrity score based on all proctoring data
 - **Factors:** Violation count/severity, tab switches, look-away count, face detection confidence, voice analysis
 - **Output:** Integrity score (0–100), review status, per-violation breakdown
 - **Priority:** High
-

3.4 Learning & Certifications

FR-LEARN-01: Learning Dashboard

- **Description:** Personalized dashboard showing training progress, assigned plans, and certification status
- **Available to:** All authenticated users
- **Priority:** High

FR-LEARN-02: Training Plans

- **Description:** Role-based training plans with topic-wise structure
- **Components:** Training plan → Topics → Learning materials (articles, videos)
- **Progress Tracking:** Per-topic progress with status (In Progress, Upcoming, Completed)
- **Actions:** Start Assessment, Practice per topic
- **Priority:** High

FR-LEARN-03: Practice Assessments

- **Description:** Non-proctored practice assessments for learning topics
- **Configuration:** Question count, difficulty, type distribution
- **Evaluation:** AI-powered feedback with topic/difficulty scores
- **Usage Tracking:** Free vs. paid tiers
- **Priority:** Medium

FR-LEARN-04: Certification Exams

- **Description:** Proctored certification exams for cloud platforms
- **Supported Providers:** Azure, AWS, GCP, Databricks, Snowflake
- **Difficulty Levels:** Beginner, Intermediate, Advanced

- **Flow:** Select certification → Pre-interview checks → Proctored exam → AI evaluation → Certificate issuance
- **Retake Policy:** Governed by `can_user_retake_certification` RPC
- **Priority:** High

FR-LEARN-05: Certificate Management

- **Description:** Certificate issuance and verification
- **Features:**
 - Certificate with unique verification code
 - PDF generation via edge function
 - Public verification page at `/verify-certificate`
 - My Certificates dashboard
- **Priority:** Medium

FR-LEARN-06: Learning History

- **Description:** Track all learning activities, assessment attempts, and feedback
 - **Pages:** `/learning-progress/:id` , `/learning-feedback/:attemptId` , `/learning-history`
 - **Priority:** Medium
-

3.5 Organization Management

FR-ORG-01: Multi-Tenant Organizations

- **Description:** Organizations as top-level entities with isolated data
- **Fields:** Name, slug, logo, industry, size, pricing model, settings, active status
- **Data Isolation:** RLS policies restrict data to organization members
- **Priority:** Critical

FR-ORG-02: Organization Member Management

- **Description:** Add, edit, remove organization members with role assignment
- **Fields:** User, role, permissions, active status
- **Invitation Flow:** Admin invites → email sent → user accepts → added to org
- **Priority:** High

FR-ORG-03: Partner Onboarding

- **Description:** New organizations apply via `/partner/onboarding`
- **Flow:** Submit application → Platform admin reviews → Approve/reject → Organization created
- **Priority:** Medium

FR-ORG-04: Organization Settings

- **Description:** Configurable organization-level settings
- **Access:** Partner admin and platform admin
- **Priority:** Medium

FR-ORG-05: Organization Analytics

- **Description:** Organization-level metrics and dashboards
 - **Metrics:** Interviews conducted, candidates evaluated, AI usage, team activity
 - **Priority:** Medium
-

3.6 Billing & Subscriptions

FR-BILL-01: Subscription Plans

- **Description:** Plan-based access control with usage limits
- **Plan Attributes:** Name, price (monthly cents), max interviews, max users, max AI usage
- **Priority:** High

FR-BILL-02: Organization Subscriptions

- **Description:** Link organizations to subscription plans
- **Integration:** Stripe for payment processing
- **Features:** Plan selection, upgrade/downgrade, usage tracking, auto-renewal
- **Priority:** High

FR-BILL-03: Invoice Management

- **Description:** Automated invoice generation and history
- **Features:** Invoice generation via edge function, line items, PDF export, payment status
- **Priority:** High

FR-BILL-04: Payment Methods

- **Description:** Manage organization payment methods
- **Storage:** Card brand, last 4 digits (sensitive data in Stripe)
- **Priority:** Medium

FR-BILL-05: Usage Tracking

- **Description:** Track interviews conducted, AI tokens used, active users per billing period
 - **Alerts:** Approaching limit notifications
 - **Priority:** Medium
-

3.7 AI Configuration & Management

FR-AI-01: AI Provider Management

- **Description:** Configure and manage AI providers
- **Providers:** Google Gemini (primary), OpenAI (optional)
- **Features:**
 - Provider registry with credentials management
 - AES-encrypted API key storage
 - Feature-to-model mapping with A/B testing
 - Feature toggles and fallback configuration
- **Priority:** High

FR-AI-02: AI Health Monitoring

- **Description:** Monitor AI service health and performance
- **Metrics:** Response time, success rate, token usage, cost per request
- **Dashboard:** Real-time health status, usage graphs, cost monitoring
- **Priority:** Medium

FR-AI-03: AI Usage Monitoring

- **Description:** Track and limit AI usage per organization
 - **Tracking:** Tokens consumed, requests made, cost per feature, latency percentiles
 - **Priority:** Medium
-

3.8 Platform Administration

FR-ADMIN-01: Platform Admin Hub

- **Description:** Central dashboard for platform administration
- **Access:** platform_admin only
- **Features:** Organization overview, user counts, system health, quick actions
- **Priority:** High

FR-ADMIN-02: Organization Management

- **Description:** Create, edit, deactivate organizations across the platform
- **Priority:** High

FR-ADMIN-03: Unified User Management

- **Description:** Manage all users across organizations
- **Features:** User list, search, role assignment, deactivation, impersonation
- **Priority:** High

FR-ADMIN-04: Partner Application Review

- **Description:** Review and approve/reject partner applications
- **Priority:** Medium

FR-ADMIN-05: Platform Configuration

- **Description:** Key-value platform settings management
- **Categories:** System config, email configuration, AI settings, payment gateways
- **Priority:** Medium

FR-ADMIN-06: Operation Logs & Monitoring

- **Description:** View and analyze platform operation logs
- **Pages:** Operation logs, log analysis (AI-powered), pre-interview check logs
- **Priority:** Medium

FR-ADMIN-07: Scheduled Jobs

- **Description:** Manage cron jobs for automated tasks
- **Jobs:** Data cleanup, deadline enforcement, stale session cleanup, reminder emails
- **Priority:** Low

3.9 Notifications & Communication

FR-NOTIF-01: In-App Notifications

- **Description:** Real-time notifications for platform events
- **Types:** New candidate submission, assessment report ready, invitation accepted, etc.
- **Features:** All/Unread tabs, Mark all read button, notification list
- **Priority:** High

FR-NOTIF-02: Email Notifications

- **Description:** Transactional emails for key events
- **Types:** Welcome, verification, invitation, reminder, password reset, review request
- **AI Enhancement:** Optional AI-powered email content enhancement
- **Customization:** Email templates with subject and body configuration
- **Priority:** High

3.10 JD Builder & Templates

FR-JD-01: JD Builder Wizard

- **Description:** AI-assisted job description creation

- **Features:**
 - Job title searchable dropdown
 - Step-by-step wizard with Next/Back navigation
 - AI-generated JD content with suggestions
 - Skill extraction from generated JD
 - Direct flow to interview creation
- **Priority:** Medium

FR-JD-02: Interview Templates

- **Description:** Reusable interview templates
- **Features:**
 - Template library with search and filters
 - Seniority filter (Junior, Mid, Senior, Lead)
 - Save interview config as template
 - Create interview from template
- **Priority:** Medium

FR-JD-03: Question Repository

- **Description:** Centralized question bank
- **Features:** Search by keyword, filter by topic, question cards with type/difficulty badges
- **Priority:** Medium

3.11 Reporting & Export

FR-RPT-01: Assessment Reports

- **Description:** Detailed per-candidate assessment reports
- **Content:** Overall score, topic scores, hiring decision, CPI breakdown, strengths/weaknesses, per-question analysis
- **Priority:** High

FR-RPT-02: Report Builder

- **Description:** Custom report generation
- **Priority:** Medium

FR-RPT-03: Data Export

- **Description:** Export data in multiple formats
- **Formats:** PDF (jsPDF), Word (docx), Excel (exceljs), HTML (html2canvas)
- **Priority:** Medium

3.12 Chatbot

FR-CHAT-01: AI Chatbot Assistant

- **Description:** Platform assistance chatbot powered by AI
- **Data Source:** chatbot_knowledge table with Q&A pairs
- **Management:** /admin/chatbot-management for knowledge base CRUD
- **Priority:** Low

4. External Interface Requirements

4.1 User Interface

Requirement	Specification
Framework	React 18.3 SPA with React Router v6
Component Library	shadcn/ui (50+ Radix primitives)
Styling	Tailwind CSS 3.4 with CSS variables theming
Responsive	Mobile-first, supports 320px–4K displays
Dark Mode	Supported via CSS class toggle
Accessibility	Radix primitives provide ARIA attributes
Code Editor	Monaco Editor for coding questions
Charts	Recharts for analytics visualizations
Diagrams	Mermaid for architecture rendering
Icons	Lucide React icon set
Toasts	Sonner toast notifications
Loading States	Skeleton loaders, PageLoader spinner
Lazy Loading	Heavy pages (proctoring, interviews) loaded lazily

4.2 Hardware Interfaces

Interface	Purpose
Camera (WebRTC)	Video recording for proctoring
Microphone (WebRTC)	Audio recording for proctoring
Screen Share (getDisplayMedia)	Screen recording for proctoring

4.3 Software Interfaces

External System	Protocol	Purpose
Google Gemini API	HTTPS/REST	AI question generation and evaluation
Stripe API	HTTPS/REST	Payment processing, subscriptions
Supabase GoTrue	HTTPS/REST	Authentication (email/password)
Supabase PostgREST	HTTPS/REST	Database CRUD via REST
Supabase Realtime	WebSocket	Real-time data subscriptions
Supabase Storage	HTTPS/REST	File/recording storage
SMTP Server	SMTP	Transactional email delivery

ATS Webhooks	HTTPS	Applicant tracking system integration
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4.4 Communication Interfaces

Protocol	Usage
HTTPS	All API communication (TLS 1.2+)
WebSocket	Real-time notifications, proctoring broadcasts
REST	CRUD operations via PostgREST
RPC	Secure database function calls

5. Non-Functional Requirements

5.1 Performance

Metric	Requirement
Page Load	First meaningful paint < 3s on 4G
API Response	95th percentile < 2s for CRUD operations
AI Generation	Question generation < 60s for 50 questions
AI Evaluation	Candidate evaluation < 30s
Concurrent Users	Support 1,000 concurrent sessions (PgBouncer)
Max Upload	Recording chunks up to 50MB each
Network Threshold	Minimum 1 Mbps for proctored sessions

5.2 Reliability & Availability

Metric	Requirement
Uptime Target	99.5% monthly
Auto-Recovery	Automatic session resume on connection loss
Retry Policy	3 attempts, 1s→5s exponential backoff (1.5× multiplier)
Retryable Errors	Network failures, HTTP 5xx, HTTP 429 (rate limit)
Background Upload	Background upload handler for interrupted transfers
Error Boundaries	React ErrorBoundary at root + per-section with Sentry reporting

5.3 Scalability

Dimension	Specification
Database Connections	PgBouncer: 1,000 max clients

Container Orchestration	Docker Compose (dev), Kubernetes + Helm (production)
Edge Functions	Stateless Deno functions, horizontally scalable
Storage	Supabase Storage API (S3-compatible backend)
CDN	Nginx serves static assets with 1-year cache headers

5.4 Security

See [Section 7: Security Requirements](#).

5.5 Usability

Requirement	Specification
Onboarding	Interactive tour + onboarding checklist for new users
Error Messages	PostgreSQL errors mapped to user-friendly messages
Toast Notifications	Consistent success/error/info toast styling
Breadcrumbs	Role-aware breadcrumbs on all protected pages
Role-Based Navigation	Navigation adapts to user's active role
Keyboard Shortcuts	Command palette for power users

5.6 Maintainability

Requirement	Specification
Type Safety	Full TypeScript across frontend
Code Quality	ESLint with custom rules
Testing	630 E2E tests (Playwright), Vitest unit tests
Logging	4-layer logging: logger, systemLogger, operationLogger, audit-logger
Error Monitoring	Sentry integration with section-tagged error boundaries

5.7 Portability

Platform	Support
Web	Chrome, Firefox, Safari, Edge (latest 2 versions)
iOS	Via Capacitor hybrid app (iOS 14+)
Android	Via Capacitor hybrid app (Android 10+)
Deployment	Docker containers, Kubernetes, AWS (migration planned)

6. Data Requirements

6.1 Database Tables (60+ tables)

Core Entities

Table	Description	Key Relations
profiles	User profiles	FK → auth.users
user_roles	Role assignments	FK → profiles
organizations	Multi-tenant orgs	—
organization_members	Org membership	FK → organizations, profiles
interviews	Interview templates	FK → organizations, profiles
questions	Question bank	FK → interviews
interview_invitations	Candidate invitations	FK → interviews
interview_attempts	Candidate submissions	FK → interviews
attempt_questions	Question-attempt junction	FK → interview_attempts, questions
assessments	AI evaluation results	FK → interview_attempts

Proctoring

Table	Description
proctoring_sessions	Recording URLs, violations, integrity score
preinterview_check_logs	Camera/mic/network/lighting status
candidate_performance_index	CPI scores per candidate

Billing

Table	Description
subscription_plans	Plan definitions
organization_subscriptions	Org-plan links
invoices	Invoice records
usage_tracking	Per-period usage
payment_gateways	Stripe configuration
payment_methods	Card details
payment_transactions	Charge records

Learning

Table	Description
learning_assessments	Practice assessments
learning_assessment_questions	Assessment questions
learning_assessment_attempts	User attempts
learning_assessment_feedback	AI feedback
training_plans	Training plan definitions
training_topics	Topics in plans
learning_materials	Materials per topic
user_training_assignments	User-plan links
user_topic_progress	Progress tracking
certification_topics	Cert definitions
certification_questions	Cert exam questions
certification_attempts	User cert attempts
user_certificates	Issued certificates

AI & Platform

Table	Description
ai_providers	Provider registry
ai_provider_credentials	Encrypted API keys
ai_model_configurations	Model-feature mapping
ai_feature_configurations	Feature toggles
ai_usage_logs	Token/cost/latency tracking
platform_configurations	Key-value settings
audit_logs	Action audit trail
security_events	Security event logs
notifications	User notifications

6.2 Data Retention

Data Type	Retention
Audit logs	90 days
Proctoring recordings	Configurable per organization

Security events	90 days
Interview attempts	Indefinite (org-managed)
Certificates	Indefinite

6.3 Data Privacy (GDPR)

- **Right to Access:** Users can view all their personal data
 - **Right to Deletion:** Account deletion removes profile, roles, and personal data
 - **Consent:** Explicit proctoring consent required before recording
 - **Encryption:** API keys encrypted with AES; passwords with bcrypt
 - **PII Protection:** Candidate email/name immutable after attempt creation
-

7. Security Requirements

7.1 Authentication Security

Requirement	Implementation
SR-01: Password hashing	bcrypt via GoTrue
SR-02: JWT session tokens	Signed JWTs with configurable expiration
SR-03: Email verification	Required before protected access (configurable)
SR-04: Auto token refresh	60-second buffer check
SR-05: HTTPS only	TLS 1.2+ enforced

7.2 Authorization Security

Requirement	Implementation
SR-06: RBAC enforcement	6 roles, 38 permissions, route/component/DB level
SR-07: Row-Level Security	406 RLS policies on all tables
SR-08: God mode isolation	platform_admin bypasses role checks (explicit)
SR-09: Organization isolation	RLS ensures org data is siloed
SR-10: Service role restricted	Service key only in edge functions, never client-exposed

7.3 Data Security

Requirement	Implementation
SR-11: API key encryption	AES encryption for AI provider credentials
SR-12: Session token validation	64-char base64 tokens, validated per request
SR-13: Answer protection	Correct answers hidden from candidates via RLS
SR-14: PII immutability	Candidate email/name locked after attempt creation

SR-15: Error sanitization	Technical errors masked to user-friendly messages
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7.4 Infrastructure Security

Requirement	Implementation
SR-16: CORS configuration	Configured on edge functions and API gateway
SR-17: Security headers	CSP, HSTS, X-Frame-Options, X-Content-Type-Options
SR-18: Non-root container	Nginx runs as non-root user
SR-19: Gzip compression	Enabled for text responses
SR-20: Static asset caching	1-year cache headers, versioned filenames

7.5 Monitoring & Audit

Requirement	Implementation
SR-21: Audit logging	All admin actions logged to <code>audit_logs</code>
SR-22: Security event logging	Security events logged to <code>security_events</code>
SR-23: Error monitoring	Sentry integration for runtime errors
SR-24: AI health monitoring	Continuous health checks on AI services

8. Appendices

8.1 Edge Functions Catalog (107 functions)

Category	Functions	Count
Interview & AI	generate-questions, evaluate-interview, generate-job-description, extract-skills, enhance-job-description, suggest-jd-content, regenerate-questions, batch-regenerate-questions, detect-bias, calculate-cpi, parse-resume	11
Assessment & Learning	evaluate-learning-assessment, evaluate-certification, generate-learning-questions, generate-certification-questions, generate-training-plan, refine-assessment-feedback	6
Proctoring	init-proctoring-session, update-proctoring-session, log-proctoring-violation, analyze-violations, analyze-proctoring-video, cleanup-proctoring-chunks, cleanup-stale-proctoring, get-proctoring-upload-url, get-chunk-upload-url, upload-proctoring-recording, upload-proctoring-screenshot, confirm-proctoring-upload, merge-proctoring-chunks, repair-webm-metadata, auto-close-sessions	15
User Management	complete-user-signup, complete-password-setup, admin-user-management, manage-organization-user	4

Notifications & Email	send-email, send-notification, send-interview-invitations, send-invitation-reminders, send-verification-email, send-password-setup, send-review-request, resend-email, auth-email-hook, enhance-email-content	10
Billing	generate-invoice, approve-partner-application, fix-pending-invitations, delete-organization	4
Documentation & Testing	generate-documentation, generate-documentation-from-code, generate-architecture-docs, update-architecture-diagram, generate-schema, run-tests, run-flow-tests, run-comprehensive-flow-test, analyze-test-error, auto-fix-issue, validate-test-data, seed-test-data, seed-flow-data, cleanup-test-data	14
Platform	chatbot-assist, scan-platform-features, scan-ai-features, test-ai-connection, test-configuration, schedule-interview, manage-scheduled-jobs, scheduled-data-cleanup, enforce-interview-deadlines, ats-webhook, sync-ats-candidates, generate-certificate-pdf	12
Other	Remaining utility functions	31

8.2 Database Functions (123 functions)

Key security-critical functions (SECURITY DEFINER):

- `has_role(uid, role) / has_any_role(uid, roles[])`
- `user_is_org_member() / user_is_org_admin()`
- `can_access_interview() / can_access_org_data()`
- `encrypt_api_key() / decrypt_api_key()`
- `generate_share_token() / generate_session_token()`
- `create_interview_attempt() / update_attempt_with_session()`
- `get_interview_for_candidate() / get_questions_for_candidate()`
- `can_user_retake_certification()`

8.3 Validation Schemas Summary

Schema	Fields	Key Rules
signUpSchema	email, password, fullName	See FR-AUTH-01
signInSchema	email, password	Email ≤255, password required
createInterviewSchema	title, jobDescription, questionCount, timeLimit, difficultyDistribution	See FR-INT-01
candidateInfoSchema	candidateName, candidateEmail	2–100 chars, letters; valid email
profileUpdateSchema	fullName	2–100 chars, letters only
answerSchema	answer	1–5,000 chars

passwordChangeSchema	newPassword, confirmPassword	Match + strength rules
roleAssignmentSchema	role	Enum of valid roles
interviewSubmissionSchema	sessionToken, answers, timeTaken	Token ≥ 32 , time 0–86,400s

End of Requirements Document